

The Audio Ease™ Rocket Science™ Bundle 3.1.0

Manual for version 3.1.0

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System requirements:

Macintosh G4/400 MHz or better
MACOS 10.2.3 or higher

Available in the following formats:

HTDM/RTAS/AudioSuite(for Pro Tools TDM&LE systems)
MAS (for Motu Digital Performer and Audio Desk)
VST (for Cubase and Nuendo & others)
Audio Unit (for Logic & others)



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An unauthorized version of the RocketScience Bundle beeps every few seconds.
To authorize your copy:

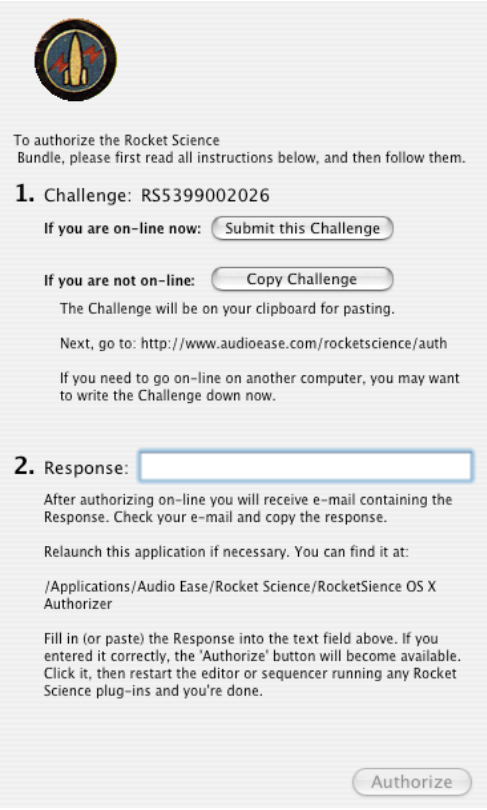
1. Install the RocketScience Bundle release version from CD or a download.

2. The installer automatically launches the RocketScience OS X authorizer.

3. In the authorizer, you see the challenge code generated for your system.

4a. If the computer you're installing RocketScience on is not connected to the internet, write down the challenge code and web address quoted and go to a computer that is connected to the internet.

4b. If the computer you're installing RocketScience on is connected to the internet, press the button labeled 'Submit this challenge'



5. On the web authorization page, follow the instructions given there.

6. If your authorization was approved, you'll receive an e-mail with the response code.

7. If the Nautilus OS X Authorizer is not running, launch it from:
/Applications/Audio Ease/Nautilus/RocketScience OS X Authorizer

8. If you have a sequencer/editor running with the RocketScience Bundle active, you'll have to quit and relaunch that to make the demo beeps go away.

You can apply human vowel sounds to your audio with Roger. Use the vowel bar at the bottom to click through the vowels and click the portrait to select one of the Roger family members.

Portamento controls the transition speed between vowels, and the 'sharpness' of the vowel is controlled by the bandwidth parameter.

Make the Roger family talk rhythmic random nonsense, locked to Performers tempo, using Auto. Roger is MIDI controllable from your MIDI keyboard or from recorded MIDI data on the sequencer's tracks, and it can receive parameter automation from Performer.

Besides a Mono to Mono Roger there is a Stereo to Stereo Roger that uses identical settings to treat both channels of stereo input.

Roger is fitted with a look ahead limiter to prevent any level accidents that can be the result of resonating filters in general.



By clicking on a face, you select the vowel set of one of the Roger Family members. Roger's mouth sits from C2 to B2 on your MIDI keyboard, Patty's vowels are from C3 to B3, and Cindy's from C4 to B4. This sequence is repeated throughout your MIDI keyboard.





This is the Vowel Bar. Click in any of these vowels to apply it to your input sound, and drag your mouse over the buttons to very quickly trigger them. You can also use a MIDI Keyboard, or paramater autoimation to trigger vowels.

Portamento is the speed at which one vowel transforms into the next. It can be controlled either by the knob, a MIDI keyboard's modulation wheel, the velocity of a MIDI keyboard parameter automation. A value of 0 will make a vowel transition so short that it will almost click. Turning the knob all the way to the right results in a transition time of about 6 seconds.





This determines the bandwidth of the formant bandpass filters that make up the vowel filter. It can be controlled either by the knob, a MIDI keyboard's modulation wheel, or the velocity of a MIDI key. A value of 0 will make your filters so narrow that they whistle (they will keep whistling when you turn of the input). Turning the knob all the way to the right will make the filters so wide that the output will in fact not be filtered at all.



Setting 'Auto' to 'on' will make any member of Roger's family babble random nonsense, in a tempo determined by the tempo knob. The babble is synced to Performers tempo, you can select multiples and divisions of your Performer bpm's. During Auto, Roger still reacts to clicks in the vowel bar, MIDI notes and parameter automation.





Much more interesting rhythms are produced by auto, if some beats are randomly skipped. Use the beat skip parameter to select what percentage of the beats should be skipped.



Tips & Hints

- Even more powerful than 'Auto' for getting a groove into Roger's babble is to us an Arpeggiator, either from the host or an external synth. It is always in sync with your song and highly customisable for weird effects. Pipe your keyboard's MIDI to Arpeggiator and feed the MIDI output of the Arpeggiator to Roger. (MOTU arpeggiator shown)

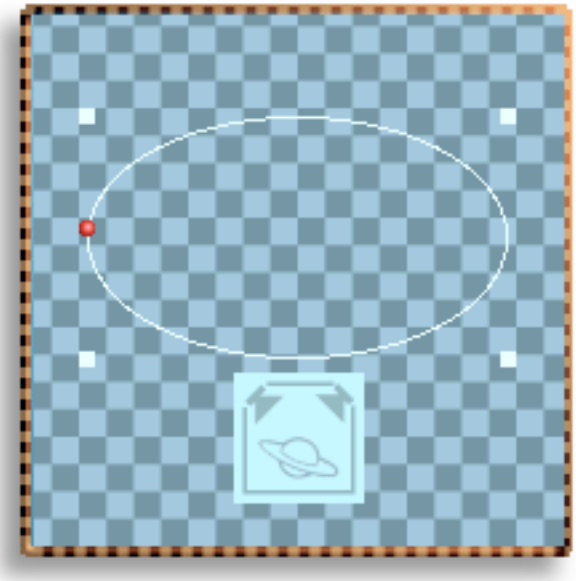
Orbit lets you control the virtual position of your sound source. It accurately calculates the sound source's reflections of the walls of a virtual 'outer room', using the exact amount of delay and attenuation for the direct sound and all reflections alike. It is a plug-in mainly of the Mono To Stereo kind, although there is a Stereo to Stereo version that mixes left and right inputs before they are sent to Orbit.

You can use Orbit merely as an automatable physically correct panning device, or you can use it as a flight path simulator that offers control over your position, speed and location of the source, the size of the virtual space and the absorption of high frequencies by the walls. For direct sound and all reflections, Orbit will precisely emulate the Doppler effect so that pitch will shift up when a sound (or a reflection) comes at you, and will shift down when it moves away.

The light blue small room is the room you are in. You can drag it up and down in the outer room, its size is determined by the speaker span parameter. The tiled surface is the outer room, that is adjusted with the room size parameter. The red ball is the sound source.

The motion selector lets you choose between a static source or one that follows a path set by clicks in the outer room. The source can also follow an adjustable oval, or it can just fly about like Buck Rogers on LSD.





This is the graphic display of the localization parameters. It is a view from above, like a road map.

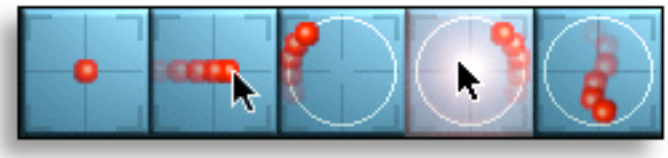
The light blue small room is the room you are in. You can drag it up and down in the outer room, its size is determined by the speaker span parameter.

The tiled surface is the outer room, that is adjusted with the room size parameter. One tile is a meter, which is about three feet.

The red ball is the sound source. Which may sit still or move, according to the motion selection at the bottom left of Orbit. Read on for more info about the white oval...



With the Motion Selector you select the motion type of the source. When you click in it the popup menu at the right comes up. It offers five choices.





- Here the source will sit still wherever you click the mouse in the outer room. It requires less processing power than the four moving types, and therefore you may be able to select more reflections. Whenever you click in a different location, you may notice an audible click. The speed knob has no effect when you select this one.



- Here the source will move to wherever you click in the outer room. When you quickly click in multiple locations the source will follow your trail, smoothly cutting some corners here and there. The maximum speed the source can reach is set by the speed knob in meters per second (check the speed knob for additional details).





- With these two, the source sound will follow the trail set by an adjustable white oval, in the clockwise or counter clockwise direction, at a cruising speed determined by the speed knob (check the speed knob for additional details).
You can change the horizontal and vertical dimensions of the oval by dragging in its white squares. You can change the location of the circle by clicking inside the circle and dragging it. You can also set the circle so narrow that it becomes a vertical or horizontal line.



- Here the source sound will move randomly about like a drunk, within the confines of the white oval, at a maximum speed determined by the speed knob (check the speed knob for additional details). Like with the above two, the circle's dimensions and location are adjustable. Try setting it small and drag it around a bit.



This is the knob that determines the (maximum) cruising speed of the source. Its unit is meters per second. One dark or light blue tile in the outer room is a meter. Your cruising speed at a stroll in the park would be about 1.5 m/s, while them Duke Boys may rush by at 40 m/s (90 mph). The speed knob is more precise at low values than at high values. You can hold option down while dragging for extra precision, or you can type in a value.



The source localization works optimally if you set the speaker span value to the distance between the speakers that you are listening to. If you set this value too high, you may experience a 'hole in the middle' effect, if you set it too low,the stereo imagery may become too narrow. Click in the Speaker Span field and drag to left and right to adjust the setting. Hold <option> (<alt>) down while dragging for more precision. In the graphic display you will see the light blue inner room size change accordingly. The minimum setting of 0.2 meter (about 8 inches) is meant for use on headphones.





The room size parameter sets the size of the outer room. In the graphics display every tile is a meter (that's approximately three feet). Click in this field and drag to left and right to adjust the setting. Hold down <option> (<alt>) while dragging for more precision.

The high damp knob determines the frequency response of the walls of the outer room. As an example: wooden walls will reflect less high frequencies than concrete ones, so to simulate them, you should apply some High Damping. When High Damp is set to zero, reflections will sound as bright as the direct sound.



Automation



All parameters of Orbit are automatable in hosts that allow plug-in authomation. Note that some parameters, for instance, changing the room size and speaker span, may cause jittering, as those parameters are not updated per sample. Doing so would severely overload your CPU.



The localization cues that the listener gets from Orbit are generated by accurately calculating the sound source's reflections of the walls of the outer room.

If you let the sound source reflect of each wall once, then you need four reflections per speaker. If you allow each of those reflections to bounce off of yet another wall, then you'll need 8 more reflections, totalling to 12 reflections. If you set the number of reflections to 60, then the sounds that reach each of the speakers of the inner room have bounced off of five walls or less. The higher the setting, the longer your reflection pattern and the more processor power Orbit will need.

Setting this number to 0 will allow only the direct, unreflected sound to reach both speakers of the inner room, therefore it will not impose the acoustic characteristics of the outer room on your sound at all, while still preserving a lot of the accuracy of the localization.

When you select the 'static' motion: (see right) the processor load is much lower than in any of the other motions. That is because in that case Orbit does not have to make smooth, click-free interpolations. Also a buffer size of 128 bytes (hardware setup) requires more processing power than a buffer size of 1024 bytes.



Tips & Hints

- Suppose you have four close miked (or MIDI'ed) instruments on separate tracks of your host. Insert an Orbit in every channel, set the number of reflections to 0 and the motion static (see above). You can now push every instrument to its own location, creating a much more realistic panorama than you could have with plain panning. The sound is still dry, so you can apply high quality (outboard) reverberation. When you let Orbit calculate early reflections, try to get rid of the early reflections your reverberation unit produces.



- To simulate the small movement an instrumentalist would make, you can select the motion to the right and confine the instrumentalist within a small circle. Consider that much of the 'panning' effect of Orbit is achieved by the difference in distance from the source to the left speaker and to the right speaker. Often you can locate a source to the far right and still maintain more or less the same volume on both channels. A trick you wouldn't be able to pull off using normal panning.

- Some parameters do not smoothly update the algorithm running, meaning they may introduce zipperlike noise. Room size, inner room position, speaker span and circle position and radius may cause this. Normally, the slower the motion of any of these parameters, the more predictable the output.

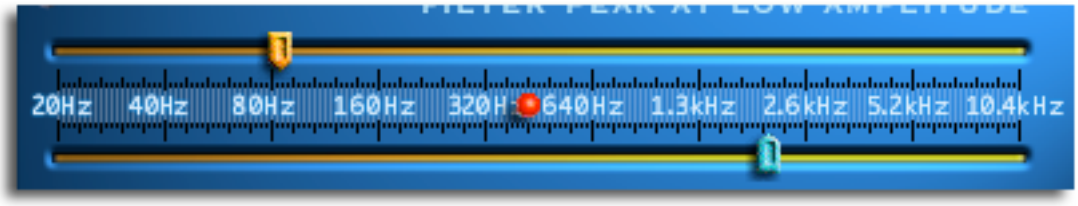
Follo™ dynamically adjusts the peak of a resonating bandpass filter according to the level of the incoming audio, and according to any incoming MIDI modulation wheel or breath control. The blue slider is the target for high amplitude audio. The moving ball indicates where the resonator's peak is, the bandwidth determines how wide and resonating the band is, and the release determines the rate at which the peak falls back to the yellow slider. The analysis knob offers control over the sensitivity of the ball. Applications of Follo include an autowah on a guitar, and a thumping resonating bass generator when the ball clips at the blue slider when it is set way low.

Besides a Mono to Mono Follo there is a Stereo to Stereo Follo that uses identical settings to treat both channels of stereo input. But you should really try the Mono to Stereo Follo, especially on sounds like (synth) basses. It can do some pretty weird things to your stereo image.

Follo is fitted with look ahead limiter to prevent any level accidents that can be the result of resonating filters in general.

All Follo's parameters are automation enabled.





The red ball indicates at what frequency the filter peak lies. That frequency will never exceed the limit of either yellow or blue slider. The direction the ball moves in is determined by the settings of the sliders relative to one another.



The yellow slider sets the frequency of the filter peak when the input signal is silence. If the yellow slider is to the left of the blue slider, the ball will move up in frequency when the input amplitude rises. If the yellow slider is to the right of the blue slider, the ball will move down in frequency when the input amplitude rises.



The blue slider sets the frequency of the filter peak when the input signal is at its max: 0dB. When the level goes up, the ball shoots towards the blue slider and when the level goes down, it falls back to the yellow one. Hold down <option> (<alt>) for more precision when adjusting the sliders.

You can adjust the level analysis sensitivity with the analysis knob. Setting it high may cause the ball to hit the blue slider often. Analysis doesn't change the output gain of the signal.





The release control adjusts the rate at which the filter will sweep back to the yellow slider frequency after a loud peak has passed.

The bandwidth control adjusts the resonant quality of the filter. A high value gently lifts a broad frequency band around the red ball, a lower value narrows the filter band, and at the lowest extreme lets the filter oscillate.



Tips & Hints

- You should really try the Mono to Stereo Follo, especially on sounds like (synth) basses. It can do some pretty weird things to your stereo image.
- Setting the analysis parameter high causes the peak to hit the blue slider often. If bandwidth is very low, this frequency will become very apparent in the resonating output. Try the effect of tuning the blue slider to the root of your music (hold down <option> (<alt>) while adjusting for more precision).
- Try inserting a delay plug-in before Follo™ in a channel. When the input sound stops, Echo with feedback makes it fade away in bumps. And Follo will follow.





Midi Control of the RocketScience Bundle.

Version 3.0 of the Rocket Science Bundle adds significant Midi control. Note that Logic does not allow effect plug-ins to be controlled by midi.

Roger:

Midi controller 1: Bandwidth and/or portamento, when set so in the user interface
Note velocity: Bandwidth and/or portamento, when set so in the user interface
Note numbers: Plays the vowels, eight in an octave, for three octaves alternating between Roger , Patty and Cindy. Notes outside the three octave range wrap around.

Orbit:

Midi controller 11: Room size
Midi controller 12: Inner room position forward and backward
Midi controller 13: Inner room left and right
Midi controller 14: Damping
Midi controller 15: Speed
Midi controller 16: Circle horizontal radius
Midi controller 17: Circle vertical radius
Midi controller 18: Circle horizontal position
Midi controller 19: Circle vertical position
Midi controller 20: Inner room size, or speaker span
Midi controller 21: Reflection order
Midi controller 1: in direct positioning modes (the ones without the circle), the back- or forward position of the sound source
Pitch bend: in direct positioning modes (the ones without the circle), the left or right position of the sound source

Patch change: Changes the movement mode, program 1 is static, 2 is following, 3 is circle clockwise, 4 is circle counter clockwise, 5 is random within circle. Higher values wrap back.

Note on: In direct positioning modes, the note number sets the horizontal position, velocity the vertical position

Follo:

Midi controller 11: Analysis, or drive
Midi controller 12: Bandwidth
Midi controller 13: Release
Midi controller 14: Low frequency slider
Midi controller 15: High frequency slider

Pro Tools notes.

You can use the Rocket Science Bundle in Pro Tools TDM system as an HTDM plug-in. This means the plug-in appears to you as regular TDM plug-in, but in the background does its DSP work on the host computer. An HTDM plug-in has some system dependent latency. With a standard Pro Tools installation, this latency is 1028 samples. You can get this latency smaller by installing the '_128' or low latency version of the Digidesign Stream Manager. Your latency will then be 260 samples. Note that the lower latency version of the Stream Manager affects all other HTDM and DirectConnect plug-ins. As always, the lower latencies you use, the higher the processor load. You have to make sure that your system is up to the job before installing the low latency version of the Stream Manager. We do not recommend the lower latency version on systems slower than 1 GHz.

The RTAS version is meant for users of Pro Tools LE systems.

As of version 3.1.0, the Rocket Science Bundle can also be used as AudioSuite plug-ins. Note that the AudioSuite plugs appear as distinct mono-mono and stereo-stereo plug-ins: you must choose the proper one for the selection you wish to process in your session or region bin.

Digital Performer notes.

There is only one specific issue with the Rocket Science Bundle and Digital Performer. As Digital Performer loads both its own MAS format plug-ins as well as Audio Units, when both versions of the Rocket Science Bundle are installed, both will show up. The Audio Unit set will be labeled Roger AU, Follo AU and Orbit AU, where the MAS versions will carry their straight names. In this way you can always make a distinction between one and the other.

In general, we recommend using the MAS version only in DP, and not have the Audio Unit installed. The Audio Unit version does not support automation, where the MAS version does.

Cubase & Nuendo notes.

There are no specific issues with running the Rocket Science Bundle in Cubase or Nuendo.

Logic notes.

To use the Rocket Science Bundle in Logic, you need to install the Audio Unit version.

New in versino 2.1.0 is the ability to automate the plug-ins' parameters. Follow the instructions in the Logic documentation on the use of automation.`

To use Midi control with the Rocket Science plug-ins, you must insert them in the instrument slot of an instrument object. Then select an input source from the sidechain menu to pass audio to the plug-ins inputs.